

Comparitive Performance Report for tentacle-ec-4+2-16k-balance-off vs direct-reads-ec-4+2-16k-balance-reads- on-rerun-pr2 vs direct-reads-ec-4+2-16k-balance-off- rerun-pr2

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Comparison summary for tentacle-ec-4+2-16k-balance-off vs direct-reads-ec-4+2-16k-balance-reads- on-rerun-pr2 vs direct-reads-ec-4+2-16k-balance-off- rerun-pr2

Sequential Read	tentacle_ec_4+2_16k_balance-off	direct-reads-ec_4+2_16k_balance-reads-on-rerun-pr2	direct-reads-ec_4+2_16k_balance-off-rerun-pr2
4K	76980@5.0ms	110994@3.5	44%
8K	76437@5.0ms	108021@3.5	41%
16K	74065@5.2ms	105114@3.6	42%
32K	60602@6.3ms	41849@9.2	-31%
64K	2751@9.1ms	1808@13.9	-34%
128K	4703@10.7ms	3392@14.8	-28%
256K	6535@15.4ms	5999@16.8	-8%
384K	7203@27.9ms	7661@26.3	6%
512K	7859@34.2ms	9330@28.8	19%
768K	8071@49.9ms	10389@38.8	29%
1024K	8390@48.0ms	11625@23.1	39%
2048K	8160@32.9ms	12026@22.3	47%
4096K	7838@68.5ms	11558@23.2	47%

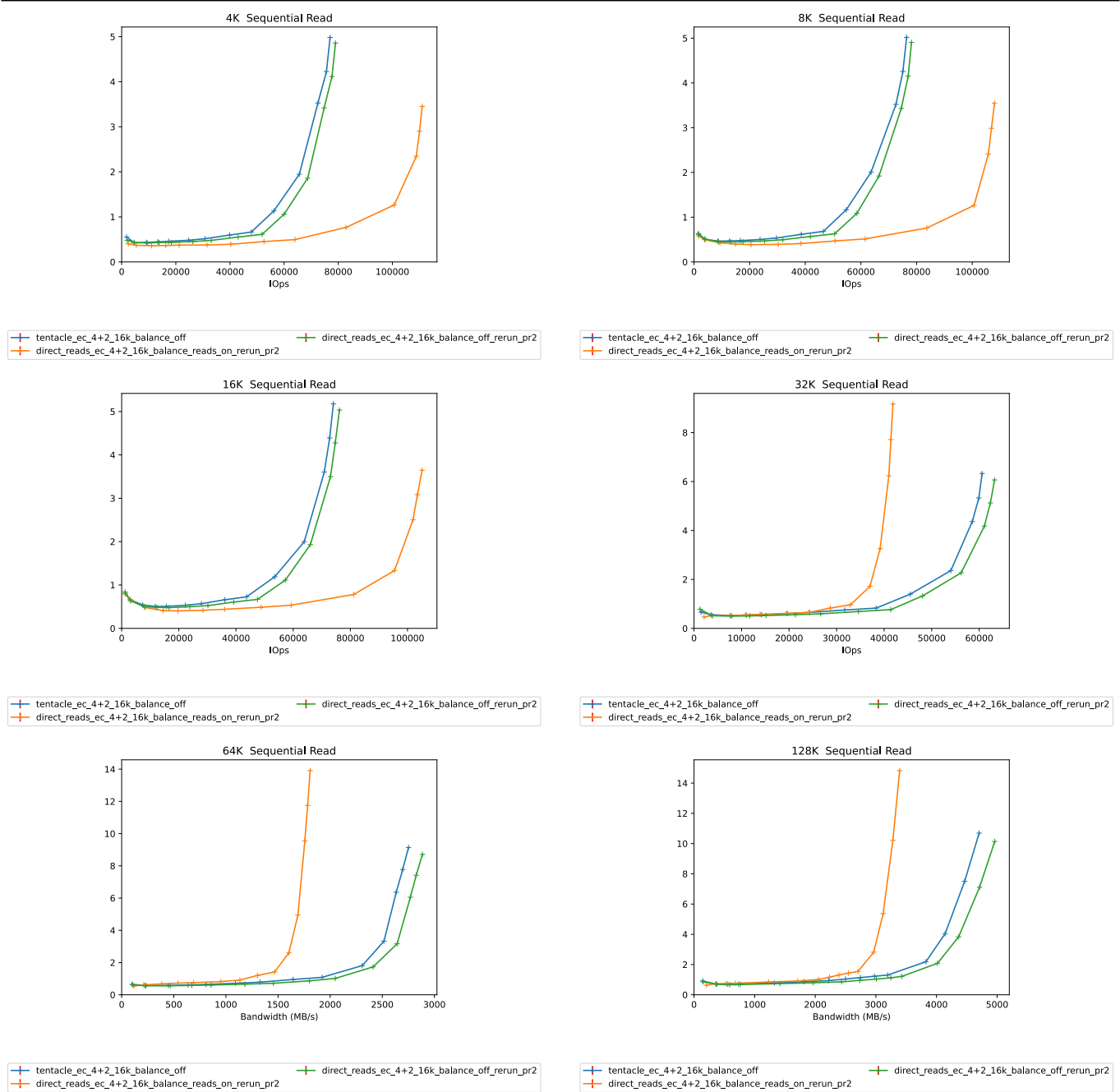
Random Read	tentacle_ec_4+2_16k_balance-off	direct-reads-ec_4+2_16k_balance-reads-on-rerun-pr2	direct-reads-ec_4+2_16k_balance-off-rerun-pr2
4K	70831@5.4ms	102355@3.7	45%
8K	69838@5.5ms	101466@3.8	45%
16K	67308@5.7ms	97685@3.9	45%
32K	48933@7.8ms	43823@8.8	-10%
64K	2117@11.9ms	1678@12.5	-21%
128K	3705@13.6ms	3058@16.5	-17%
256K	5897@17.1ms	5165@19.5	-12%
384K	6543@30.8ms	6560@30.7	0%
512K	7079@37.9ms	7940@33.8	12%
768K	7452@54.0ms	9093@5.5	22%
1024K	7794@51.7ms	10516@6.4	35%
2048K	7951@16.9ms	11729@11.4	48%
4096K	7768@69.1ms	11570@11.6	49%

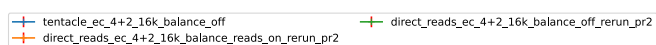
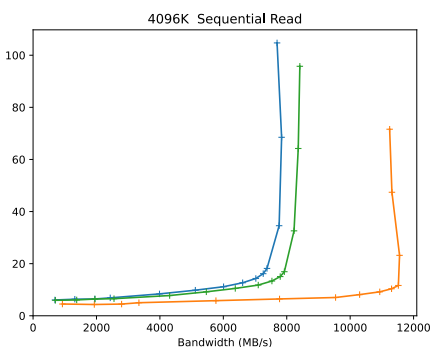
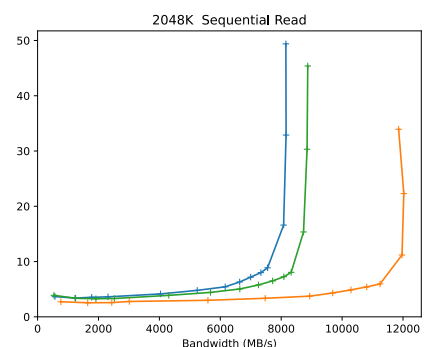
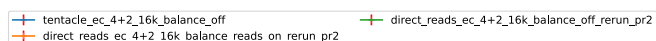
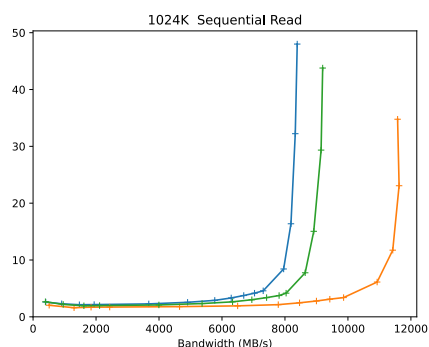
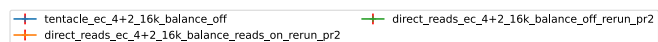
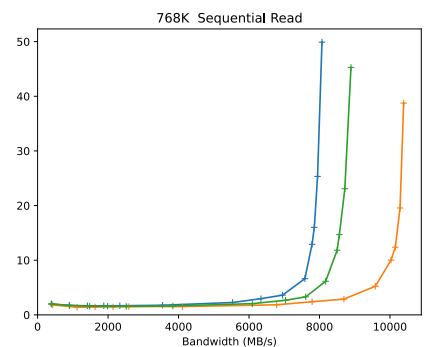
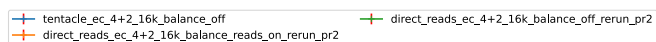
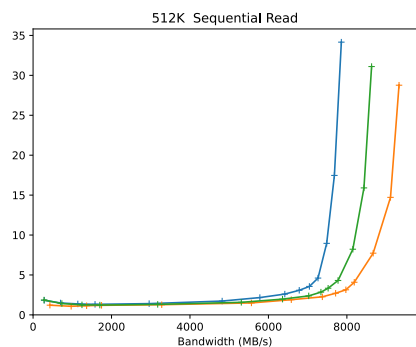
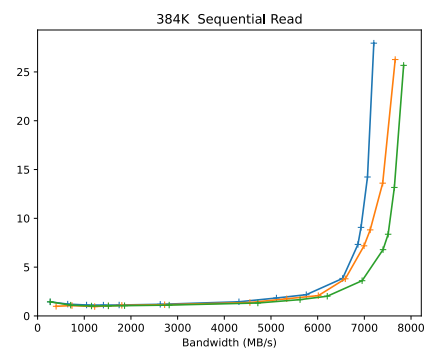
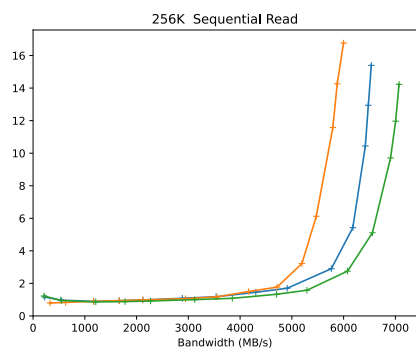
Random Read/Write	tentacle_ec_4+2_16k_balance-off	direct-reads-ec_4+2_16k_balance-reads-on-rerun-pr2	direct-reads-ec_4+2_16k_balance-off-rerun-pr2
4K_70/30	33776@11.4ms	39676@9.7	17%
16K_70/30	32052@10.0ms	37723@8.5	18%

Random					
Read/Write	tentacle_ec_4+2_16k_balance-off	direct-reads-ec-4+2_16k_balance-off	direct-reads-ec-4+2_16k_balance-reads-on-rerun-pr2	direct-reads-ec-4+2_16k_balance-off-rerun-pr2	
64K_70/30	1397@15.0ms	1182@17.7	-15%	1443@14.5	3%
64K_30/70	1065@23.6ms	998@25.2	-6%	1106@22.7	4%

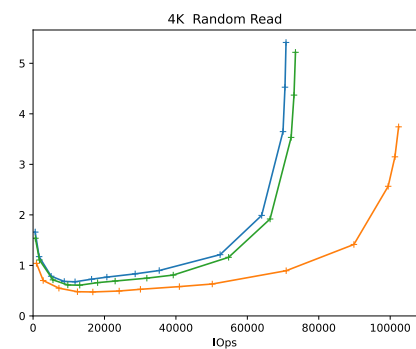
Response Curves

Sequential Read

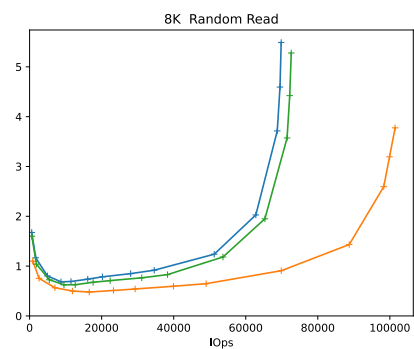




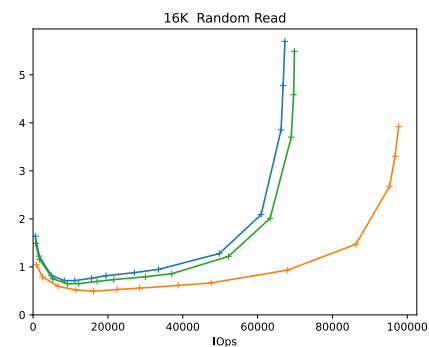
Random Read



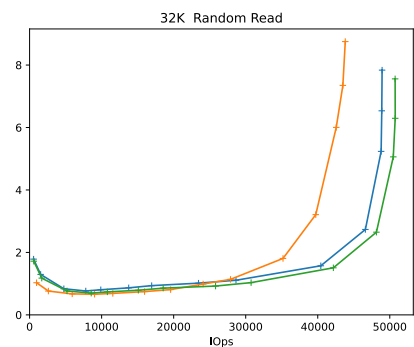
tentacle_ec_4+2_16k_balance_off direct_reads_ec_4+2_16k_balance_off_rerun_pr2
direct_reads_ec_4+2_16k_balance_reads_on_rerun_pr2



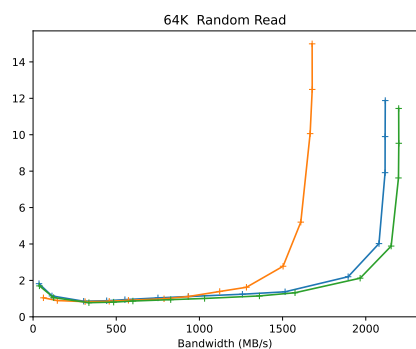
tentacle_ec_4+2_16k_balance_off direct_reads_ec_4+2_16k_balance_off_rerun_pr2
direct_reads_ec_4+2_16k_balance_reads_on_rerun_pr2



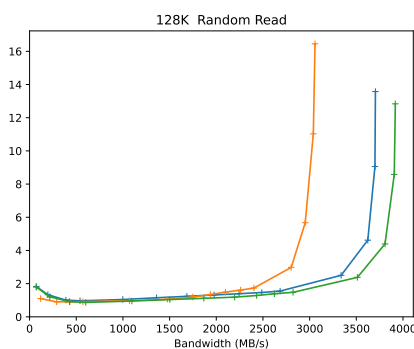
tentacle_ec_4+2_16k_balance_off direct_reads_ec_4+2_16k_balance_off_rerun_pr2
direct_reads_ec_4+2_16k_balance_reads_on_rerun_pr2



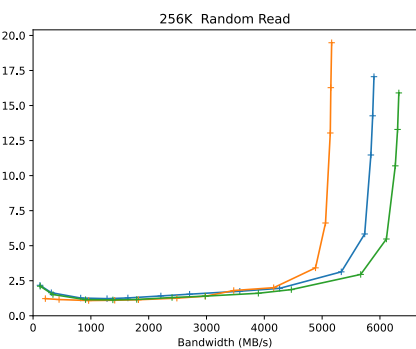
tentacle_ec_4+2_16k_balance_off direct_reads_ec_4+2_16k_balance_off_rerun_pr2
direct_reads_ec_4+2_16k_balance_reads_on_rerun_pr2



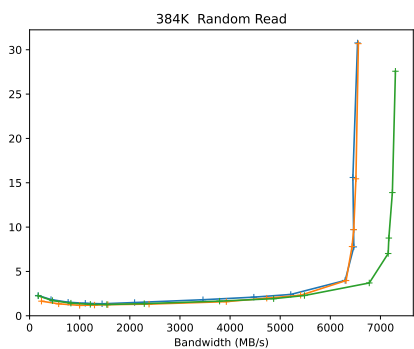
tentacle_ec_4+2_16k_balance_off direct_reads_ec_4+2_16k_balance_off_rerun_pr2
direct_reads_ec_4+2_16k_balance_reads_on_rerun_pr2



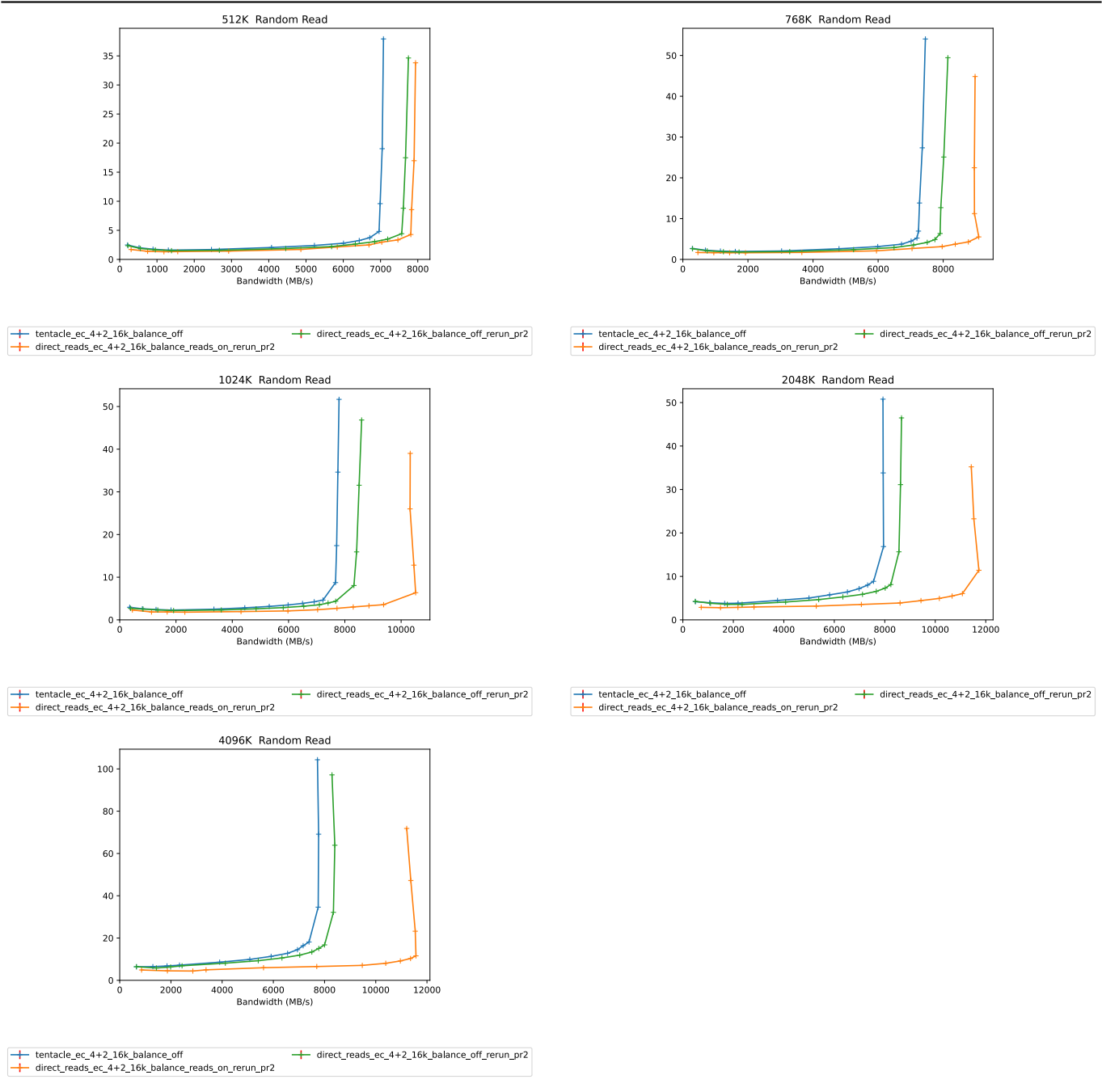
tentacle_ec_4+2_16k_balance_off direct_reads_ec_4+2_16k_balance_off_rerun_pr2
direct_reads_ec_4+2_16k_balance_reads_on_rerun_pr2



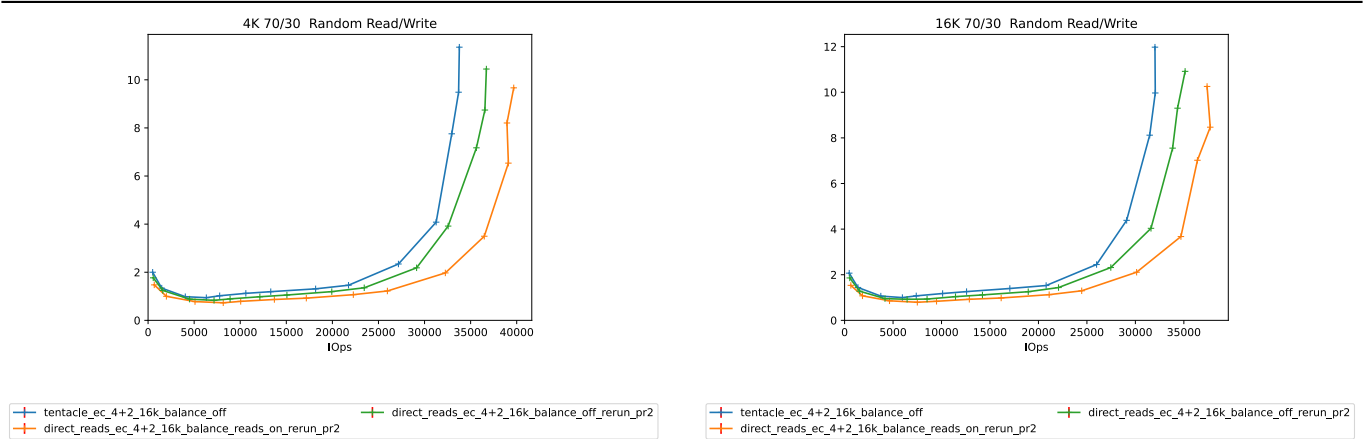
tentacle_ec_4+2_16k_balance_off direct_reads_ec_4+2_16k_balance_off_rerun_pr2
direct_reads_ec_4+2_16k_balance_reads_on_rerun_pr2

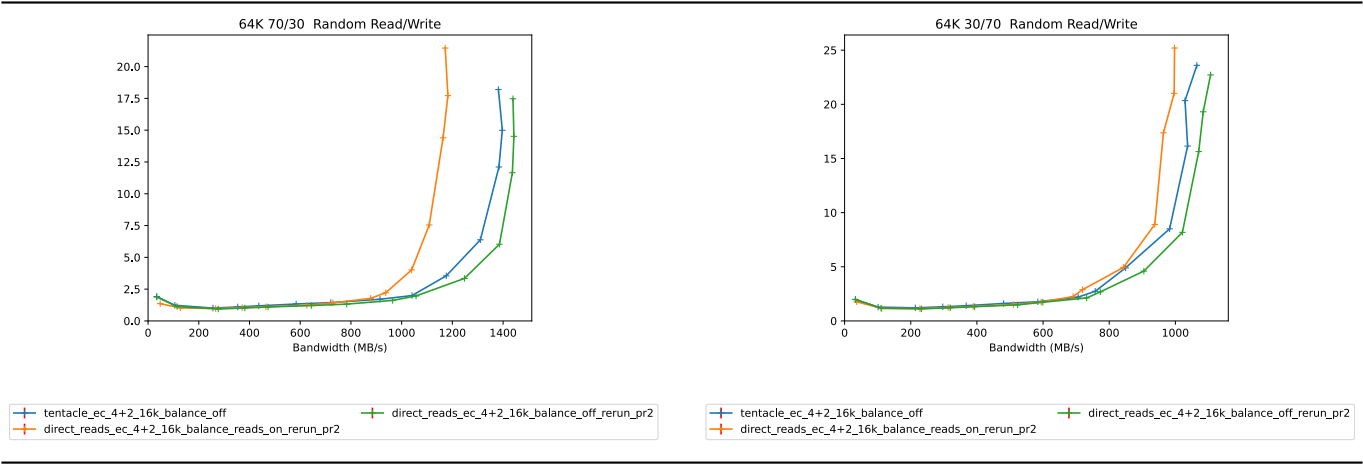


tentacle_ec_4+2_16k_balance_off direct_reads_ec_4+2_16k_balance_off_rerun_pr2
direct_reads_ec_4+2_16k_balance_reads_on_rerun_pr2



Random Read/Write





Configuration yaml files

Only yaml files that differ by more than 20 lines from the yaml file for the baseline directory will be added here in addition to the baseline yaml

results

```

librbd fio:
  cmd_path: /usr/local/bin/fio2
  create_report: true
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
    - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
    - 1
  ramp: 30
  rbdname: cbt-librbd fio
  time: 90
  time_based: true
  use_existing_volumes: true
  vol_size: 1000
  volumes_per_client:
    - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    128k randomread:
      jobname: randread
      mode: randread
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8
        - 12
        - 16
        - 20
        - 24
        - 28
        - 32
        - 64
        - 128
        - 256
        - 384
    128k sequentialread:
      jobname: seqread
      mode: read
      numjobs:
        - 1
      op_size: 131072
      total_iodepth:
        - 1
        - 2
        - 3
        - 4
        - 8

```

```
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 256
- 384
16kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 16384
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
16ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 16384
  total_iodepth:
    - 1
    - 2
    - 4
```

```

- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
1Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 1048576
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 256
    - 384
256krandomread:
  jobname: randomread
  mode: randread
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1

```

```

- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
256ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 262144
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
2Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
2Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 2097152
  total_iodepth:
    - 1

```

```
- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192
32krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
32ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 32768
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
384krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1
```

```

- 2
- 3
- 4
- 5
- 8
- 16
- 24
- 32
- 64
- 128
- 160
- 256
- 512
384ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 393216
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 160
    - 256
    - 512
4Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
    - 64
    - 128
    - 192
4Msequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4194304
  total_iodepth:
    - 1

```



```

- 2
- 3
- 4
- 8
- 12
- 16
- 20
- 24
- 28
- 32
- 64
- 128
- 192
4kmixread70:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 4096
  rwmixread: 70
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
4krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 4096
  total_iodepth:
    - 1
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 320
    - 384
4ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 4096
  total_iodepth:

```

```

- 1
- 2
- 4
- 6
- 8
- 12
- 16
- 24
- 32
- 64
- 128
- 256
- 320
- 384
512krandomread:
  jobname: randread
  mode: randread
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
512ksequentialread:
  jobname: seqread
  mode: read
  numjobs:
    - 1
  op_size: 524288
  total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
    - 128
    - 256
    - 512
64kmixread30:
  jobname: mixread
  mode: randrw
  numjobs:
    - 1
  op_size: 65536
  rwmixread: 30

```

```

    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  64kmixread70:
    jobname: mixread
    mode: randrw
    numjobs:
      - 1
    op_size: 65536
    rwmixread: 70
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  64krandomread:
    jobname: randread
    mode: randread
    numjobs:
      - 1
    op_size: 65536
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  64ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1

```

```

    op_size: 65536
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  768krandomread:
    jobname: randread
    mode: randread
    numjobs:
      - 1
    op_size: 786432
    total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 8
      - 16
      - 24
      - 32
      - 40
      - 48
      - 64
      - 128
      - 256
      - 512
  768ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1
    op_size: 786432
    total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 5
      - 8
      - 16
      - 24
      - 32
      - 64
      - 128
      - 160
      - 256
      - 512
  8krandomread:
    jobname: randread
    mode: randread
    numjobs:
      - 1

```

```

    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  8ksequentialread:
    jobname: seqread
    mode: read
    numjobs:
      - 1
    op_size: 8192
    total_iodepth:
      - 1
      - 2
      - 4
      - 6
      - 8
      - 12
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
      - 320
      - 384
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
      - 1
    op_size: 65536
    time: 600
    total_iodepth:
      - 16
cluster:
  archive_dir: /tmp/cbt
  ceph-mgr_cmd: /usr/bin/ceph-mgr
  ceph-mon_cmd: /usr/bin/ceph-mon
  ceph-osd_cmd: /usr/bin/ceph-osd
  ceph-run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
    - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
    --- server1 ---:

```

```

    a: null
mkfs_opts: -f -i size=2048
mons:
  --- server1 ---:
    a: --- IP Address ---:6789
mount_opts: -o inode64,noatime,logbsize=256k
osds:
- --- server1 ---
osds_per_node: 6
pdsh_ssh_args: -a -x -l%u %h
rados_cmd: /usr/bin/rados
rbd_cmd: /usr/bin/rbd
tmp_dir: /tmp/cbt
use_existing: true
user: cjames
monitoring_profiles:
  collectl:
    args: -c 18 -sCD -i 10 -P -oz -FO --rawtoo --sep ";" -f {collectl_dir}

```